

AI Crop Disease Diagnosis Report

Generated on 2026-07-09 17:17:31

Crop Name:	Rice
Location:	Guntur , Andhra Pradesh
Symptoms:	yellow spots
Diagnosis Result:	Rice Brown Spot

Weather Information:	34.5°C, Humidity: 72%, Wind: 12.4 km/h (Partly cloudy)
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Analyzed Leaf Image:



Diagnostic Details & Farming Advisory

Dear Valued Farmer,

We understand your concerns regarding the yellow spots observed on your rice crop in Guntur, Andhra Pradesh. Based on the symptoms reported and our AI Vision analysis, we have identified a specific issue affecting your field. This report provides detailed guidance to help you manage and mitigate the disease effectively, ensuring the health and productivity of your crop.

Disease Name

Common Name: Rice Brown Spot

Scientific Name: *Bipolaris oryzae* (syn. *Helminthosporium oryzae*)

Disease Description

Rice Brown Spot is a prevalent fungal disease that significantly impacts rice yield and quality. It manifests as numerous small to medium-sized, oval to irregular lesions primarily on the leaves, but can also affect leaf sheaths, panicles, glumes, and grains. The spots typically appear dark brown to reddish-brown, often exhibiting a lighter, yellowish-brown or greyish center with a distinct darker margin. Our AI vision observed these lesions, noting some were coalescing, leading to general yellowing and drying on affected leaves, consistent with advanced infection stages. This reduces the plant's ability to photosynthesize, weakening it and potentially leading to fewer and lighter grains, ultimately impacting your harvest.

Possible Causes

Rice Brown Spot is caused by the fungus *Bipolaris oryzae*. This disease thrives under specific environmental conditions, which are highly relevant to your current situation in Guntur:

- Environmental Factors:** The current weather in Guntur, with a temperature of 34.5°C and 72% humidity, provides an ideal environment for the proliferation of this fungal pathogen. High humidity and warm temperatures promote spore germination and fungal growth. Partly cloudy conditions can also contribute to prolonged leaf wetness, further aiding infection.
- Nutritional Stress:** Plants under stress, particularly those deficient in essential nutrients like potassium, phosphorus, calcium, magnesium, or silicon, are more susceptible to Brown Spot. Imbalanced fertilization, especially excessive nitrogen application without adequate other nutrients, can also predispose plants to infection.
- Inoculum Presence:** The fungus can survive on infected crop debris in the field or on weed hosts, serving as a source of inoculum for subsequent crops.

Recommended Pesticides

For effective chemical control of Rice Brown Spot, fungicides can be applied. It's crucial to apply them at the first sign of symptoms and ensure thorough coverage.

- Active Ingredients:** Look for fungicides containing active ingredients such as:
 - Propiconazole:** Apply at 1 mL per liter of water.
 - Azoxystrobin:** Apply at 1 mL per liter of water.
 - Mancozeb:** Apply at 2-2.5 grams per liter of water.
- Spray Frequency:** A single application might be sufficient if the disease is caught early and conditions improve. However, if conditions remain favorable for the disease (high humidity, warm temperatures) or symptoms persist, a second application might be necessary after 10-14 days.

Always follow the manufacturer's instructions for specific dilution rates and safety precautions.

Organic Treatment Methods

For farmers preferring eco-friendly approaches, several organic methods can help manage Rice Brown Spot:

- Neem Oil:** A natural fungicide and insecticide, neem oil can be effective. Prepare a 1-2% solution (10-20 mL of neem oil per liter of water) with an emulsifier (like mild soap) and spray thoroughly. Repeat every 7-10 days if needed.
- Bio-fungicides:** Products containing beneficial microbes like *Trichoderma harzianum* or *Pseudomonas fluorescens* can help suppress fungal pathogens and promote plant health. *Pseudomonas fluorescens* can also be used as a seed treatment (10 g/kg seed) to provide early protection.
- Cow Dung Extract:** Spraying fresh cow dung extract (20%) can improve plant vigor and potentially offer some protective effect against pathogens.
- Field Sanitation:** Remove and destroy infected crop residues after harvest to reduce the fungal inoculum for the next season. Avoid leaving infected plant material in the field.

Fertilizer Suggestions

Balanced nutrition is key to building crop resistance against diseases like Brown Spot. * **Reduce Nitrogen:** While nitrogen is essential for growth, excessive application makes plants lush and more susceptible to fungal diseases. Apply nitrogen in split doses according to soil test recommendations. * **Increase Potash (Potassium):** Potassium is vital for strengthening plant cell walls and enhancing overall disease resistance. Ensure adequate potassium application, especially during the tillering and panicle initiation stages. * **Consider Silicon:** Silicon application can also significantly increase the plant's defense mechanisms against various stresses, including fungal diseases. * **Soil Testing:** Regularly conduct soil tests to understand your field's nutritional status and apply fertilizers precisely.

Irrigation Recommendations

Proper water management is crucial for minimizing disease spread and plant stress. * **Avoid Deep Flooding:** While rice is a wetland crop, prolonged deep flooding can stress plants and create favorable humid microclimates for fungal diseases. Ensure moderate water levels. * **Optimal Drainage:** Ensure your fields have good drainage systems, especially during heavy rains. This helps prevent waterlogging around the plant roots, which can contribute to stress and disease susceptibility. * **Intermittent Drying:** Consider practicing intermittent irrigation where feasible, allowing the field to dry out briefly between irrigation cycles. This can help reduce humidity levels around the crop canopy and disrupt the disease cycle. * **Avoid Water Stress:** Ensure plants receive adequate water, especially during critical growth stages, to prevent drought stress that can weaken plants and make them more vulnerable.

Preventive Measures

Implementing preventive measures is the most sustainable way to manage diseases in the long term. * **Resistant Varieties:** Choose rice varieties known to be resistant or tolerant to Brown Spot. Consult with local agricultural extension services for recommendations suitable for Guntur. * **Seed Treatment:** Treat seeds with fungicides (e.g., Carbendazim + Thiram) or bio-fungicides (**Pseudomonas fluorescens**) before sowing to eliminate seed-borne inoculum and provide early protection. * **Proper Spacing:** Ensure optimal plant spacing to allow for good air circulation within the canopy, which helps reduce humidity and leaf wetness, making it less favorable for fungal growth. * **Crop Rotation:** Rotate rice with non-host crops where possible. This helps break the disease cycle and reduces the buildup of fungal spores in the soil. * **Weed Control:** Keep fields and bunds free of weeds, as some weeds can act as alternative hosts for the pathogen. * **Field Sanitation:** After harvest, remove and destroy all crop residues, as these can harbor fungal spores and provide an inoculum source for the next season. Allow fields to dry completely during fallow periods.

By implementing these recommendations promptly and consistently, you can effectively manage Rice Brown Spot and protect your valuable rice crop. Should you observe any changes or require further assistance, please do not hesitate to contact your local agricultural extension officer.

Sincerely,

Your Expert Agricultural Consultant